

Asteroid Deflection Simulator

User Manual

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1 Introduction

The **Asteroid Deflection Simulator** is a Python-based GUI application for modeling asteroid deflection missions using ion beam spacecraft. The simulator allows users to:

- Select or edit asteroid, thruster, and spacecraft profiles.
- Define mission parameters and run MATLAB-based orbital simulations.
- Save and load scenarios.
- Visualize simulation results to show orbital changes over time.

2 Installation Requirements

- Python 3.8 or higher
- Required Python packages:
 - PyQt5
 - NumPy
 - SciPy
 - Matplotlib
- MATLAB installed and accessible via command line
- Required MATLAB files: `ODE_Handle_GUI.m`, `propagate_WOT.m`, `propagate_WT.m`, `orbit_elements.m`, `get_horizons_rv.m`
- Additional modules: `utils.py`, `utils_gui.py`, `editors.py`

3 Launching the Application

Run the GUI from the terminal or command line:

```
python gui_main.py
```

Upon launching, the main window displays dropdown menus for asteroid, thruster, and spacecraft profiles, as well as input fields for mission parameters.

4 User Interface Overview

Asteroid Deflection Simulator

Asteroid Profile:
2024 PDC25
Edit Asteroid Parameters

Thruster Profile:
Next-C Ion Thruster
Edit Thruster Parameters

Spacecraft Profile:
Single Array and Craft
Edit Spacecraft Arrangement

Scenario:
Test 1

Asteroid Mass (M_A , kg): 6410000000.0
Asteroid Radius (R_A , m): 75
Horizons ID: 2024 PDC25
Horizons Start Date (e.g. 2032-Jul-06): 2032-07-05
Horizons Stop Date (e.g. 2032-Jul-07): 2032-07-06
Horizons Step Size (e.g. 1d): 1d
Thrust (N): 0.235
Isp (s): 4178
Array Number (int): 3
Number of SC (int): 2
Mass fuel per SC (kg): 2203.0
Mass spacecraft (kg): 5867.0
Time step dt (s): 3600
Standoff distance d (m): 1000
Ion beam divergence theta (deg): 5
Orbit window (deg): 180
Infront (1 or -1): 1

Run MATLAB Simulation Save Scenario

Ready.

Figure 1: Main GUI window of the Asteroid Deflection Simulator.

4.1 Profile Selection

- **Asteroid Profile:** Choose from default or user-defined asteroid data.
- **Thruster Profile:** Select ion thruster configuration.
- **Spacecraft Profile:** Choose the number and configuration of spacecraft.
- **Scenario Selector:** Load saved scenarios with predefined parameter sets.

4.2 Editing Profiles

Buttons are provided for each profile type:

- **Edit Asteroid Parameters**
- **Edit Thruster Parameters**
- **Edit Spacecraft Arrangement**

Default profiles are read-only, while custom profiles are editable. Editors update the corresponding JSON files.

The image shows three separate application windows for editing profiles. Each window has a title bar with a question mark and a close button. The first window, 'Edit Asteroid: 2024 PDC25', contains input fields for Mass (kg), Radius (m), Horizons ID, Horizons Start Date, Horizons Stop Date, and Horizons Step Size, along with 'Save Changes', 'Save As New Profile', and 'Delete Profile' buttons. The second window, 'Edit Thruster: N...', contains input fields for Thrust (N) and Isp (s), along with the same three buttons. The third window, 'Edit Spacecraft Arrangement:...', contains input fields for Array Number, Number of Spacecraft, Mass Fuel per Craft (kg), and Spacecraft Dry Mass (kg), along with the same three buttons.

Figure 2: Asteroid Editor

Figure 3: Thruster Editor

Figure 4: Spacecraft Editor

4.3 Mission Parameter Inputs

- Asteroid: mass (kg), radius (m), Horizons ID
- Thruster: thrust (N), specific impulse (s)
- Spacecraft: number of arrays, number of spacecraft, fuel mass, spacecraft mass
- Mission: time step (s), standoff distance (m), ion beam divergence (deg), orbit window (deg), infront parameter (1 or -1)

5 Saving and Loading Scenarios

1. Input all desired parameters.
2. Click **Save Scenario**.
3. Enter a scenario name in the popup dialog.
4. The scenario is stored in `scenarios.json` and can be reloaded from the scenario dropdown.

6 Running Simulations

1. Ensure Horizons ID, start date, stop date, and step size are filled.
2. Click **Run MATLAB Simulation**.
3. MATLAB will be invoked to compute the initial asteroid state and propagate the ODE for deflection.
4. The GUI locks until the simulation completes.

6.1 Simulation Output

After completion, two outputs are presented:

- **Interactive Plot:** Displays semi-major axis, eccentricity, and B-plane deflection over time.

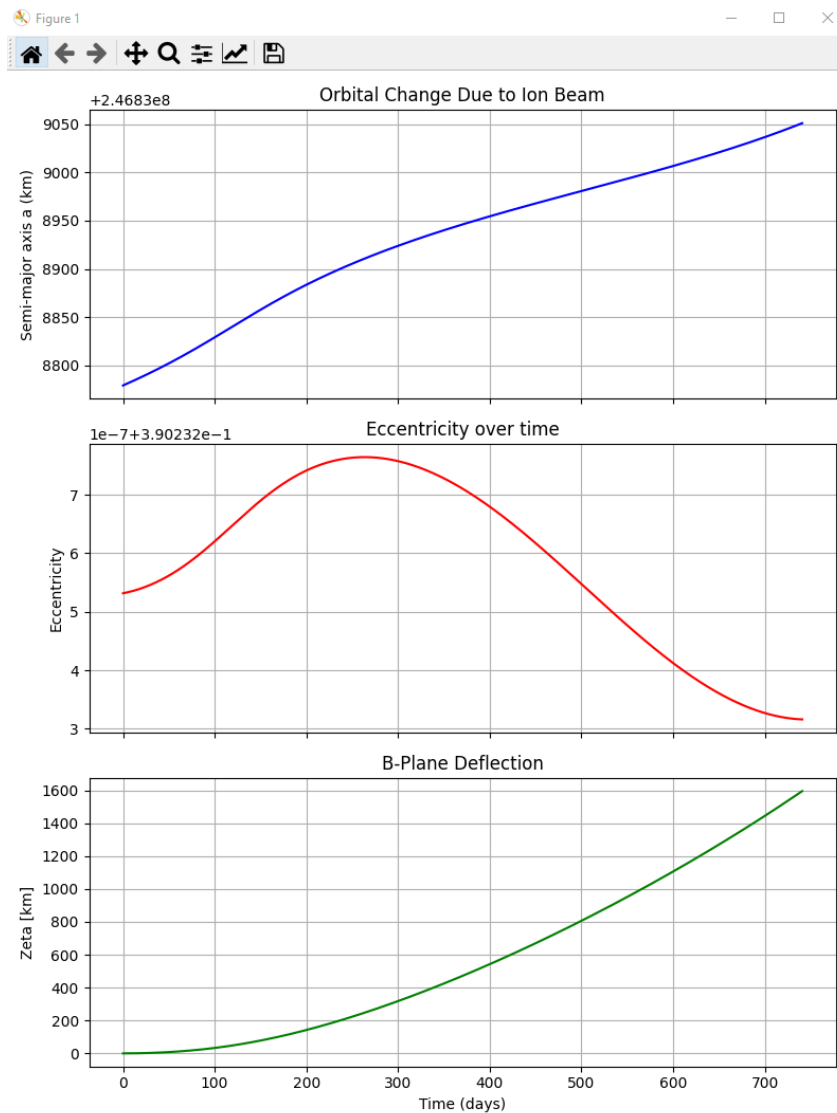


Figure 5: Subplots showing orbital change over time

- **Summary Popup:** Shows key metrics:
 - Change in semi-major axis
 - Total deflection (zeta)
 - Total delta-V imparted
 - Total change in period
 - Total simulation time in days and years

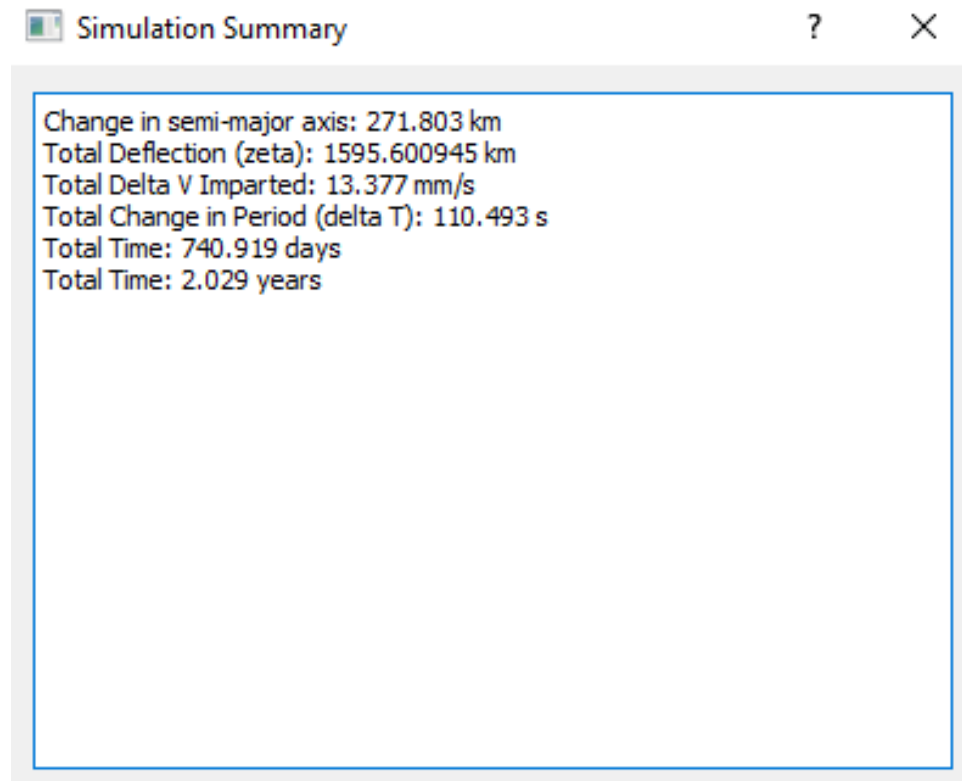


Figure 6: Simulation Summary

7 File Management

All files are stored within a JSON file inside a folder named, "data"

- `asteroids.json` - Stores asteroid profiles
- `thrusters.json` - Stores thruster profiles
- `spacecraft.json` - Stores spacecraft arrangements
- `scenarios.json` - Stores saved scenarios

8 Example Usage

1. Select asteroid “2024 PDC25”.
2. Choose thruster “Next-C Ion Thruster”.
3. Select “Single Array and Craft” spacecraft.
4. Set time step to 3600 s, standoff distance to 500 m, and orbit window to 180 deg.
5. Save scenario as “Test Run”.
6. Run simulation and observe plots and summary.

9 Troubleshooting

- **MATLAB not found:** Ensure MATLAB is installed and accessible via command line.
- **Simulation fails:** Check Horizons ID and date inputs.
- **GUI freezes:** Simulation is synchronous with GUI. It will run if given enough time (at most 1-2 minutes), even if OS states the program is not responding.

10 References

- PyQt5 Documentation: <https://www.riverbankcomputing.com/software/pyqt/intro>
- MATLAB Command-Line Interface: https://www.mathworks.com/help/matlab/matlab_external/run-matlab-functions-from-the-command-line.html